Narrating Thought and Perception*

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Abstract Good storytellers reveal their characters' 'inner stories', composed of subjective experiences like feelings, perceptions, and thoughts. In fiction, narrators can faithfully represent their main characters' thoughts, using techniques like free indirect discourse. But perceptions are fundamentally different from thoughts in that they don't have a linguistic form for the narrator to reproduce or mimic in a report. We argue that there is in fact a dedicated linguistic category, narrated perception, similar to but distinct from free indirect discourse, that allows narrators to vividly represent perceptual experiences. We propose a discourse semantics analysis of this phenomenon, built on the idea that narrators combine narrated perception and free indirect discourse into coherent discourse segments to provide a more complete representation of a fictional character's complex mental life.

keywords: narrative, discourse structure, free indirect discourse, represented perception, complex mental states

1 Introduction

Read the following passage from a Dutch children's novel:

(1) Rosa nudged him and looked around. The fly was buzzing around Mr. Gabriel's desk. She had already seen one in the dining room, at breakfast. And when she went to the toilet she was followed. They were watching her, that was clear. Hector knew that she had found the fly. But he didn't know what she had learned that night. Right? She sighed. She had to do something. Tomorrow was the day. Tomorrow Hector would kidnap her mother. The general would seize power and start a war. Then everything would be over. She had to take a chance. She had to tell someone. Rosa looked at her teacher. He was writing. The fly was buzzing above his notebook. Mr. Gabriel waved it away, but the fly kept coming back. Apparently Hector wanted to know what he wrote. Suddenly she knew for sure. Mr. Gabriel was not in on the plot. Why else would Hector spy on him?

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Mr. Gabriel was smart and he knew a lot. Maybe he could help her?
[Translation of Jozua Douglas, *De Gruwelijke Generaal*, 2015]

The narration is in the third person. The first sentence describes two events that happened in the story world: Rosa nudged someone and looked around. Reading on, we start to notice various linguistic elements that seem to represent the private thoughts and subjective experiences of the main character, Rosa, as she experiences them. Take the phrase 'that was clear' – it's not the narrator who informs us that something became clear to them, but Rosa, reasoning towards the realization that she's being watched (by fly-shaped drones). Similarly, the hesitant rhetorical question 'Right?' is not posed by the narrator, but by Rosa (to herself, as part of her internal reasoning process). Finally, the combination of a future indexical 'tomorrow' with a past tense 'was' doesn't even make sense if we consider this to be a statement from the narrator. These are all familiar signs of a form of narrative perspective shift known as free indirect discourse, a technique linguistically characterized as a mixture of direct discourse (expressives, indexicals, speech acts, and modals are interpreted relative to the perspective of the character) and indirect discourse (tenses and pronouns follow the narrator's perspective), but (typically) without overt report or quotation marking (Schlenker 2004, Eckardt 2014, Maier 2015b). From a literary point of view, it's a powerful stylistic tool for granting the reader access to the mental life of a character, vividly representing the flow of thought, from their perspective (Banfield 1982, Fludernik 1993).

But where does the free indirect discourse start?

(2) a. Rosa nudged him and looked around.
b. The fly was buzzing around Mr. Gabriel's desk.
c. She had already seen one in the dining room, at breakfast.
d. And when she went to the toilet she was followed.
e. They were watching her, that was clear.

The first, (2a), is clearly what we call plain narration, i.e., the narrator letting us know what is happening in the story world. The last, (2e), does not describe an objective fictional fact, but rather Rosa's thought ("They're watching me, that's clear"), represented in the form of a free indirect discourse. In between, (2b) appears to invoke Rosa's perspective in the specific address form 'Mr. Gabriel', especially in the Dutch original, 'meester Gabriel', which is characteristic of a student referring to her teacher. But it does not really describe a conscious thought, i.e., Rosa doesn't literally say to herself "The fly is buzzing around Mr. Gabriel's desk". Subtle forms of perspective shift like this, which are not clearly identifiable as either free indirect discourse or plain narration, have been studied under various names and subsumed under different theoretical concepts:[1] focalization (Genette 1980), character focus (Stokke 2021), protagonist projection (Stokke 2013, Abrusán 2021), viewpoint shift (Hinterwimmer 2017), (representation of) nonreflective consciousness (Banfield 1982), represented perception (Brinton 1980), and narrated perception (Fludernik 1993, Pallarés-García 2012). We follow the last terminology for examples like (2b) in the context of (2a): Given the context of the looking event set up in (2a), (2b) is to be interpreted as representing the content of Rosa's perception.

What makes this phenomenon interesting is that a perceptual experience itself has no compositional, language-like form – arguably unlike an occurrent thought, which we'll model as a form of inner speech. But it does have content, and so the narrator can try to approximate the character's subjective experience in words. This conceptual point forms the starting point of this paper. Focusing on reports of perception, our approach differs from other semantic analyses of narrative perspective shifting beyond free indirect discourse, such as Stokke's (2021) account of character focus, Stokke's (2013) and Abrusán's (2021) accounts of protagonist projection, and Hinterwimmer's (2017) of viewpoint shift. Thus, while some examples will overlap, the phenomena do not coincide. Examples like (3), for example, which these authors discuss on a par with examples that qualify as narrated perception, are beyond the scope of our analysis:

(3) a. *Everyone knew that stress caused ulcers*, before two Australian doctors in the early 80s proved that ulcers are actually caused by bacterial infection. [Hazlett 2010, cited by Abrusán 2021]

b. I wanted to be home in case he came back early, made it in time ... The house was empty, and I dove into bed, fell back asleep, and when he [Toph] came back home *his brother was there, of course had been there the whole time, of course had never left*. [D. Eggers, *A Heartbreaking Work of Staggering Genius*, 2000, cited by Hinterwimmer 2017]

In this paper we intend to capture an intuitive reading of the phenomena exemplified by the Rosa passage: (2a) describes Rosa looking (in plain narration); (2b) describes her subjective visual experience (in narrated perception); which then triggers the conscious thought process described in the subsequent sentences (in free

1 An anonymous reviewer and Banfield (1982) mention that some scholars have subsumed such phenomena under a more general header of free indirect discourse. Banfield cites Cohn (1978: 133ff) in this respect, but Cohn's position is somewhat ambiguous. In any case, like Banfield, we argue against such a unification, and for a separation of narrated perception and free indirect discourse on conceptual and linguistic grounds, and hence use the term 'free indirect discourse' in a more restricted sense, also in line with Fehr (1938), Brinton (1980) and Pallarés-García (2012), among others.

indirect discourse).

We use a notation with unmarked text for plain narration, green for represented perception, and blue for free indirect discourse. We occasionally **highlight in bold** the most relevant linguistic features, like perception verbs and perspective-sensitive elements that provide linguistic evidence for our analysis. To illustrate the phenomena in question, and our notation, here is our annotation of the entire Rosa passage:

(4) Rosa nudged him and **looked** around. The fly was buzzing around **Mr.** Gabriel's desk. She had already seen one in the dining room, at breakfast. And when she went to the toilet she was followed. They were watching her, **that was clear**. Hector knew that she had found the fly. But he didn't know what she had learned that night. **Right?** She sighed. She **had to** do something. **Tomorrow was** the day. Tomorrow Hector would kidnap her mother. The general would seize power and start a war. Then everything would be over. She **had to** take a chance. She **had to** tell someone. Rosa **looked** at her teacher. He was writing. The fly was buzzing above his notebook. **Mr.** Gabriel waved it away, but the fly kept coming back. **Apparently** Hector wanted to know what he was writing. Suddenly she knew for sure. Mr. Gabriel was not in on the plot. **Why** else would Hector spy on him**?** **Mr.** Gabriel was smart and he knew a lot. **Maybe he could** help her**?**

In Sections 2 and 3 we establish narrated perception as a genuine linguistic category. We formulate concrete linguistic characteristics that can be used to tease it apart from both plain narration (Section 2) and free indirect discourse (Section 3). In Section 4 we work out a complication concerning vocabulary choice. Section 5 elaborates on the idea, observed above for (4) already, that a description of a character's perceptions can form a bridge to a description of their thoughts. We work towards a coherence-driven approach where representations of perception and thought combine to form complex 'mental stories', narrating different aspects of the mental life of a fictional character. In Sections 6 and 7 we show how to integrate this idea into a general formal discourse semantic framework by combining a theory of discourse coherence (SDRT, Asher & Lascarides 2003) and a theory of mental representation (MSDRT, Kamp 2015).

2 Narrated perception vs. plain narration

2.1 Narrated perception represents the character's mental experience

Narrated perception is a linguistic perspective-taking technique used to represent a character's sensory perception – visual, auditory, olfactory, tactile – without explicitly reporting an act of perception in the sentence itself ('she saw that he was ill', 'he saw her approaching') (Cohn 1978, Fludernik 1993, Pallarés-García 2012). (5) gives an example:

(5) He **looked** at his mother. Her blue eyes were watching the cathedral quietly.
[D.H. Lawrence, *Sons and Lovers*, 1961, cited by Brinton 1980, 371]

Like free indirect discourse, narrated perception is syntactically free – there is no embedding or other dedicated overt marker. As a result, it is typically impossible to say of a single sentence in isolation whether it constitutes narrated perception or plain narration, or indeed free indirect discourse. We have to take the discourse context into account. Doing that we see that narrated perception is often announced in preceding sentences by the use of 'perception indicators' (Fehr 1938), often perception-related verbs like 'looked' in (5) and 'saw' in (6) and (7):

(6) He **saw** one of the men who had returned with Silva. He was standing in his boat and he was wearing a pair of mosquito-boots.
[V.S. Pritchett, *Dead Man Leading*, 1937, cited by Fehr 1938, 101]

(7) He flipped it [the light switch], but he **saw** only a dim light respond from the distance, enough to illuminate the narrow hallway before him. Yellow-painted brick walls marked with graffiti were on each side of him.
[Elizabeth Strout, *Anything is Possible*, 2017]

In all of these the narrated perception sentences are meant to describe not just states of affairs in the (fictional) world, but the character's perceptions thereof. We can turn this into a linguistic diagnostic for teasing apart narrated perception from plain narration in multiple ways.

First, a narrated perception can often be accurately paraphrased with an overt experience verb:[2]

(8) He looked at his mother. He saw her blue eyes watching the cathedral quietly.

Another linguistic consequence of the reportative nature of a passage in narrated perception is that it can never be followed by a sentence conveying that the character was unaware of the content of the narrated perception.

2 Some perception verb paraphrases work better than others for this purpose. As we'll see below, narrated perception is non-veridical. Therefore, the non-factive construction with a participle (as also used in (8)) is often a more faithful paraphrase than a finite *that*-clause complement which is generally considered factive. In addition, *that*-clause complements after perception verbs can have metaphorical, non-perceptual interpretations ('she saw that it was futile' ≈ 'she realized that it was futile'). See, e.g., Dik & Hengeveld (1991) for discussion of the differences among perception-verb complements.

(9) He looked at his mother. Her blue eyes were watching the cathedral quietly. # But he didn't notice it.

Note that (9) is perfectly well-formed and we can still make sense of it, but the second sentence is not conveying the same information as in the original. The only available interpretation for (9) treats the second sentence as a neutral description of a fact by the narrator. Given the clear perception indicator ('looked at') in the first sentence, this plain narration reading forced by the continuation in (9) leads to a confusing garden path effect, hence the hashtag.

A third linguistic consequence of the fact that narrated perception reports a sensory experience is non-veridicality. Our senses tend to be fairly reliable sources of information about the external world, so as readers we can typically deduce from the fact that he saw her watching the cathedral, or the fact that she heard a fly buzzing, that she *was* watching the cathedral and there *was* a fly buzzing. But our senses may deceive us, and narrated perception can be used to report such hallucinatory or distorted, non-veridical perceptions.

(10) Mary stepped out of the boat. The ground was shaking beneath her feet for a couple of seconds. [adapted from Hinterwimmer 2017]

(11) Sam woke up and listened carefully. There was a sharp ringing sound. Where was this annoying noise coming from? [constructed]

(10) has a prominent reading on which the ground did not really shake, but Mary just had that feeling as a result of her sense of balance being disturbed by the boat trip (Hinterwimmer 2017). Likewise, (11) could be a description of an experience of someone with tinnitus, in which case there is no ringing soundwaves out in the world, only Sam's mental experience.

In (12) we read about a particular kind of bug, aphids, hopping up and down on a rug:

(12) Jerry put down the dog, which was wrapped in a towel, and knelt over the shag rug. "I'll show you one," he said. The rug was covered with aphids; they hopped up everywhere, up and down, some higher than others. He searched for an especially large one, because of the difficulty people had seeing them. [Philip K. Dick, *A Scanner Darkly*, 1977]

On the face of it, this could be plain narration. The direct context already suggests, however, that these bugs are only perceived by Jerry, the protagonist: we read that other people have difficulty seeing them. The larger context provides further evidence that we have to do with a hallucination: the doctor has told Jerry there are no bugs in his hair, other people don't feel the bugs, and, eventually, it becomes clear that Jerry is addicted to a psychoactive drug.

Beyond hallucinations, more subtle instances of non-veridicality involve subjective descriptions that don't really correspond to the way things are objectively.

(13) He **stared at** what seemed a careless daub of paint, then stepped away. Immediately all the beauty flashed back into the canvas.

[Jack London, *Martin Eden*, 1909]

As a matter of objective fact, the beauty did not immediately flash into a canvas – this is just a description of a subjective perceptual experience.

We can paraphrase the non-veridicality criterion in terms of commitment. The narrator, by definition, is committed to the truth of the contents expressed in plain narration – even in cases of unreliable narration, where the narrator may be lying or confused. In narrated perception, by contrast, the narrator is not committed to the truth of the contents described.

Summing up: Narrated perception reports a perceptual experience but is syntactically unmarked as such. This gives us the following linguistic heuristics for distinguishing between plain narration and narrated perception:

(i) Narrated perception can be paraphrased with an overt perception verb (without changing the truth conditions).

(ii) Narrated perception cannot be consistently followed by a denial of the relevant character perceiving (e.g. 'but she didn't notice').

(iii) Narrated perception is non-veridical. The narrator is not committed to the described eventualities actually occurring.

2.2 Narrated perception represents the character's perspective

Narrated perception constitutes a form of perspective shift. The reported perception is presented from the point of view of the experiencing character. Now, perspective and viewpoint are notoriously difficult concepts, especially in the context of narrative, and we won't give a precise definition here. Instead, we gather under this heading a number of phenomena discussed in the previous linguistics literature that show that a correct interpretation of the narrated perception requires taking into account the subjective point of view of the character, not just that of the narrator.

The clearest indicator of perspective shift would be a shifted indexical. We've already seen an example of shifted 'tomorrow' constituting strong evidence for perspective shift in free indirect discourse in (1). Unfortunately, you can't see or otherwise perceive *tomorrow*, so we don't expect to find it in a perception report. Other alleged indexicals that we might in principle expect to find in perception descriptions, are not so clearly indexical (e.g., 'here' or 'now', see Hunter 2013, 2012), or are the very indexicals that we don't expect to shift in narratives anyway (pronouns and tenses, see Schlenker 2004, Maier 2017).

Looking beyond Kaplan's list of pure indexicals, we find cases of narrated perception with more subtly perspectival or deictic expressions such as 'come' vs. 'go', which indeed take the character as the deictic anchoring point:

(14) Sara got up and went to the window. A crowd had gathered outside the public house. A man was being thrown out. There he **came**, staggering.
[Virginia Woolf, *The Years*, 1937, cited by Banfield 1982]

'Came' is perspectival verb that by default indicates movement towards the speaker. In (14), it indicates a man moving in the direction of the character Sara. In other words, 'come' is interpreted relative to the character's context.

Similarly, 'left' in (15) is relative to the character, Psmith:

(15) Looking towards the window, Psmith saw that they were nearing the park. The great white mass of the Plaza Hotel showed up on the left.
[P.G. Wodehouse, *Psmith, Journalist*, 1915]

And a shifted perspective involves more than just a shift to the location and time of the perceiving character. Take Hinterwimmer's T-Rex example.

(16) [The T-Rex] hesitated. Maybe the little dinosaurs had hidden themselves in the cave on his left. When Billy **looked up** in his hiding place a few seconds later, **a T-Rex** bent down to the entrance of the cave and squinted into the dark.
[Hinterwimmer 2017]

If one only considered the common ground between narrator and reader, we would expect in the final sentence a definite description ('the T-Rex') to refer back to the previously introduced and highly salient T-Rex. The indefinite is licensed here because it suits the perspective of the character Billy.

These examples show that, in narrated perception, the narrator does not just select what the character experiences, but represents it in a way that is faithful to their perspective.

3 Narrated perception vs. free indirect discourse

3.1 The role of reflection

So far we have seen features of narrated perception that set it apart from plain narration and group it together with some other forms of mental state reporting, especially free indirect discourse. Like narrated perception, free indirect discourse describes the character's mental experience; doesn't involve overt embedding under a mental state verb; is non-veridical; and involves perspective shift. (17) gives an example:

(17) The hair was curled, and the maid sent away, and Emma sat down to think and to be miserable. – It was a wretched business, indeed! – Such an overthrow of everything she had been wishing for. – Such a development of everything most unwelcome! – Such a blow for Harriet! – That was the worst of all. [Jane Austen, *Emma*, 1815, cited by Eckardt 2014]

The sentences in blue are descriptions of Emma's thoughts. Applying the diagnostics from Section 2, note for instance that we cannot consistently add "But these thoughts never crossed her mind" and that the contents of her thoughts may not have been accurate; Emma may, for example, be wrong about how bad it was for Harriet. Likewise, we see the representation of the character's viewpoint in the choice of perspective-sensitive expressions, such as 'tomorrow' in (18), an excerpt of (1), which is to be interpreted from Rosa's perspective:

(18) Tomorrow was the day. Tomorrow Hector would kidnap her mother.

So, what's the difference between the two techniques?

From (17) we infer that Emma is having a thought of the form given by the words, including the characteristic word choices, phrasings and exclamatives (Banfield 1982, Schlenker 2004, Maier 2015b). The narrator faithfully reproduces the thoughts going through the character's mind, adjusting only the tenses (from present 'is' to past 'was' and from present perfect 'have been' to pluperfect 'had been') and pronouns (from first person 'I' to third person 'she'). In other words, free indirect discourse presents the character's consciousness as having a linguistic form (Maier 2015b). This is crucially different in the examples of narrated perception that we have seen in Section 2, if only because we don't usually think of our visual perceptual experiences as linguistic. Take (5), repeated here as (19):

(19) He **looked** at his mother. Her blue eyes were watching the cathedral quietly. [D.H. Lawrence, *Sons and Lovers*, 1961, cited by Brinton 1980, 371]

The second sentence is a description of what the character sees, but we don't assume that he formulates his experience in this (or any) linguistic form to himself. His perception is *unreflective*, in contrast to the *reflective* thoughts of Emma in (17).

To use Russell's (1940) example (cited also by Banfield 1981): you may notice that there is a puddle and act on it by stepping aside without any conscious reflection. Later, when someone asks for your reasons to step aside, you can reflect on your experience and put it into a linguistic form ('I saw that puddle and stepped aside to avoid it'). In the narratological literature this distinction goes under the name of 'reflective' vs. 'non-reflective' consciousness (Kuroda 1979, Banfield 1981).

Reflection is the defining distinction between thoughts on the one hand and perceptions and sensations on the other. In addition, we model reflection as mental verbalization. Thoughts are reflective mental processes having a linguistic form, while perceptions and sensations are unreflective processes that have no linguistic form.[3]

It is important to keep in mind that this is just folk psychology: it may well be that the distinction is difficult to maintain neuroscientifically or philosophically, for instance because what we perceive is influenced by what we believe (a phenomenon known as cognitive penetration, see e.g. Siegel 2012). But proper appreciating of 19th century literature, or contemporary fanfiction, should not depend on the latest insights from philosophy of mind and neuroscience. What matters to us, is how authors and consumers of fictional narratives naturally and intuitively conceptualize the distinction between thought and perception.

The distinction between reflective and unreflective mental states underlies the distinction between free indirect discourse and narrated perception. Free indirect discourse is a way of vividly and faithfully reporting thoughts, by quoting (salient aspects of) their form. Narrated perception is a way of vividly and faithfully reporting perceptions. Since perceptions have no linguistic form, such a linguistic report cannot be a literal quotation. The narrator instead must represent the character's experience some other way, paraphrasing the content in words that mimic as closely as possible the perspective of the experiencer (see Section 2.2).

3.2 Linguistic markers of reflection

Drawing the distinction between free indirect discourse and narrated perception in terms of reflective vs. unreflective mental processes has linguistic consequences that we can use to tease the two types of constructions apart. Reflective thought is inner speech, having roughly the same expressive power as outer speech. We may silently state facts, but also ask ourselves questions, argue with ourselves, or use expressive vocabulary like swear words and exclamatives. A narrator who has access

3 Abrusán (2021) treats her distinction between free indirect discourse and protagonist projection likewise in terms of whether there is the presumption of a (silent or loud) speech act (free indirect discourse) or not (protagonist projection). One important difference between the analyses is that for us free indirect discourse essentially includes a representation of the *form* of that speech act, as we will work out in the following sections, whereas Abrusán, despite her observation of the many quotation-like aspects in free indirect discourse, treats both free indirect discourse and protagonist projection at the level of content, following Eckardt's (2014) analysis of free indirect discourse. For a comparison, see Bary (2025).

to the relevant character's thoughts has the option of faithfully quoting all these colorful features of inner speech in free indirect discourse. Unreflective perceptual experiences on the other hand are not linguistically structured and the expressive power of such non-verbal mental representation is limited in various ways.

Below we will go through various linguistic categories that we can thus use to tease apart free indirect discourse and narrated perception.

Speech acts

While thoughts can have interrogative force, we cannot see with interrogative force, nor hear exclamatively. It follows that, as Banfield (1982) already noted, linguistic surface features of such non-assertoric speech acts, such as question marks, exclamatives, and vocatives are incompatible with narrated perception. Thus, in Section 1, we used the question marks and wh-words as evidence for identifying the following fragments of the Rosa passage (1) as free indirect discourse:

(20) Suddenly she knew for sure. Mr. Gabriel was not in on the plot. **Why** else would Hector spy on him**?** Mr. Gabriel was smart and he knew a lot. Maybe he could help her**?**

Crucially, even in descriptions of a visual experience, we consider question marks and wh-words symptoms of language-like reflection and hence as an indicator of free indirect discourse rather than narrated perception:

(21) - but **what** was she looking at**?** At a man pasting a bill.

[Virginia Woolf, *To the Lighthouse*, 1927, cited by Brinton 1980]

Modals and attitudes

Modal expressions, broadly construed including attitude verbs, counterfactuals, and other intensional constructions, indicate that we're dealing with a thought rather than a perception report, since one cannot perceive modal properties, i.e. properties that rely on what's happening in other possible worlds. We've already relied on modal verbs and adverbs in identifying free indirect discourse in our initial reading of the Rosa passage in Section 1.

(22) She **had to** do something. Tomorrow was the day. Tomorrow Hector would kidnap her mother. The general would seize power and start a war. Then everything would be over. She **had to** take a chance. She **had to** tell someone. Rosa looked at her teacher. He was writing. The fly was buzzing above his notebook. Mr. Gabriel waved it away, but the fly kept coming

back. **Apparently** Hector **wanted to know** what he was writing. Suddenly she knew for sure. Mr. Gabriel was not in on the plot. Why else would Hector spy on him? Mr. Gabriel was smart and he knew a lot. **Maybe he could** help her?

We have the evidential expression 'apparently', multiple uses of the deontic verb 'had to', the attitude verb 'wanted' (combined with epistemic 'know') and in the last sentence the combination of epistemic modal 'maybe' and ability modal 'could'.

Tense

Past and future states and events, like states and events in other possible worlds, cannot be perceived and thus belong to the realm of reflective thought (Pallarés-García 2012: 175, see also Chafe 1994). In our reading of the Rosa fragment in (23), for example, the description of the fly's buzzing is narrated perception, but the description of what Rosa had seen earlier that day, partly in the pluperfect, is a thought report:

(23) Rosa nudged him and looked around. The fly was buzzing around Mr. Gabriel's desk. She had already seen one in the dining room, at breakfast. And when she went to the toilet she was followed.

Put differently, we can only perceive what is going on right now. Tense use in a narrated perception must track the surrounding narrative tense. This makes past tense the natural candidate for narrated perceptions in past tense stories, and present tense for stories written in the (historical) present. Free indirect discourse, however, has no such restrictions as we can have thoughts about past, present and future states and events alike.

Miscellaneous

Occasionally, a character's perceptual impression is elaborated with an explicit comparison, or simile. This comparing involves an act of reflection and thus, on our view and contra Brinton (1980), similes are incompatible with narrated perception and may thus mark a shift from narrated perception to free indirect discourse.

(24) Vicky sat, her almost-all-white hair cut short, with a fringe around it, **as though it had been cut with a bowl on her head**;

[Elizabeth Strout, *Anything is possible*, 2017]

It is worth noting that Banfield's (1982, Ch.5) project is very similar to ours here, viz. establishing both an epistemological distinction between reflective and non-reflective consciousness, and an accompanying linguistic distinction between free indirect discourse and narrated perception. In addition to speech act markers like exclamatives and direct questions, discussed above, she adds parenthetical constructions as incompatible with narrated perception but compatible with free indirect discourse. Thus, despite the explicit perception verb, (25) only has a reflective, non-perceptual reading, i.e., one where 'she saw' is metaphorically interpreted as 'she realized'.[4]

(25) Paul could choose the lesser in place of the higher, she saw. [(D.H. Lawrence, *Sons and Lovers*), 1913, cited by Banfield 1982]

4 Complication: narrated perception shows the character's linguistic persona

So far, we have charted a number of linguistic features that set narrated perception apart from plain narration on the one hand and from free indirect discourse on the other. There is one last characteristic feature often associated with narrated perception, however, and this feature might appear puzzling and seem to blur the distinction between narrated perception and free indirect discourse. As Fludernik (1993) and other literary scholars stress, narrated perception, like free indirect discourse, can contain various traces of the character's idiosyncratic word choices, style, or 'linguistic persona.'[5]

For instance, in (26), 'papa' indicates the perspective of the son, as no one else would likely be in a position to use that particular address form to refer to this man.

(26) He looked at his father. Papa stood tall with his arms crossed and nodded his confirmation. [reddit.com/r/writingprompts]

Similarly, in the description of Rosa's perception in (27) we had already seen a man named Gabriel referred to as 'meester' in the Dutch original (lit. 'master'), which is a title reserved for schoolchildren to address their teacher.

(27) Rosa gaf hem een por en keek rond. De vlieg zoemde rond het bureau van **meester Gabriel**.
Rosa nudged him and looked around. The fly was buzzing around Mr.

4 As pointed out by an anomymous reviewer, Banfield even suggests a linguistic marker that can only occur in narrated perception, viz. the use of proper names to refer to the attitude holder. Although this would perhaps strengthen our case further, we don't use it as a conclusive criterion, especially in light of Maier's (2015b) examples of proper names in free indirect discourse.

5 We use 'linguistic persona' to refer to the way we create a persona, or 'mask' (i.e., the outward appearance of our identity), through our linguistic behavior. It is a central term in sociolinguistic research where linguistic variation is viewed as a resource deployed by agentive speakers and listeners to signal social identity (e.g., Podesva 2007, Burnett 2019).

Gabriel's desk.

This Mr. Gabriel may of course have many other roles in his life (be someone's neighbor, father etc.), many of which will even be unknown to Rosa. The narrator chooses the expression that fits with Rosa's word choice.

In (28) we see a definite description ('the lovers') chosen to reflect the wording that characterizes Emma's view on the situation.

(28) For ten minutes she [Emma] could hear nothing but herself. It could be protracted no longer. She was then obliged to be finished and make her appearance. **The lovers** were standing together at one of the windows. [Jane Austen, *Emma*, 1815, cited by Pallarés-García 2012: 180]

This passage occurs at a point where reader and narrator both know that the intended referents, Harriet and Mr. Elton, are not in fact lovers, as Mr. Elton is in love with Emma. The narrator's word choice thus serves to reflect an aspect of Emma's state of mind.

And in (29) we find the expression 'the girl' which clearly reflects the character Tommy's perspective. In this case we even find a narratorial comment explaining that this way of referring to the salesclerk is actually incorrect.

(29) He put the book back on the display and went to find the salesclerk to ask about a book on gardening. "I might have just the thing, we *just* got this in," and **the girl** — who was not a girl, really, except they all seemed like girls to Tommy these days — brought him a book with hyacinths on the cover, and he said, "Oh, that's perfect." **The girl** asked if he wanted it wrapped, and he said Yes, that would be great [Elizabeth Strout, *Anything is Possible*, 2017]

So, narrated perception may reflect aspects of the character's linguistic persona by mimicking their characteristic choice of words or phrasing. This goes beyond mere perspective shift, as discussed in Section 2.2. Perspective shift could be captured by a monstrous operator shifting some semantic parameters to change the interpretation of a context-sensitive lexical item. The relevant lexical items discussed here ('papa','meester', 'the lovers', 'the girl') are not indexicals whose interpretation is shifted to the reported character. What we see here is words being lifted from the character's lexical inventory. In other words, narrated perception, like free indirect discourse, involves not just context shift but also 'language shift' (Maier 2015b, Hess & Bary 2020).[6] This is different from Abrusán's (2021) account

6 Without 'language shift' we would expect that synonyms, with the same extension in all possible worlds, can be used interchangeably in narrated perception. But when we think of words that differ only in register, like 'urinate' vs. 'pee' vs. 'piss', we see that they cannot.

where both free indirect discourse and protagonist projection involve just context shift.

The issue is this: narrated perception doesn't have a linguistic form. So what does it mean exactly to say that narrated perception reflects or quotes the character's vocabulary? It cannot be the vocabulary that the character actually used, since no words were used – not even mentally.

To address this we appropriate Stokke's (2021) analysis of the different (but related) literary perspective shifting techniques of 'character focus' and 'protagonist projection': it's not the words that the character did use, but that they *would* (or, as Stokke has it, *could*) use. More specifically, we propose the following counterfactual quotation analysis: narrated perception gives the words that the character would use if they reflected on their perceptions.

We can think of this counterfactual in terms of possible worlds: the character did not (in the fictional base-worlds) use any words, since perception is non-verbal, but in the closest worlds where they mentally token some words to reflect on what they saw or heard, they use these words. By contrast, free indirect discourse is used to report the words the character in fact uttered (in thought, or out loud) in the fictional base-worlds.

Applied to (27), the narrator reports Rosa's perception of the buzzing fly with the words "The fly was buzzing around Mr. Gabriel's desk", thereby conveying that she has a perceptual experience that is such that if she used words to reflect on her perception, she would use these, including the characteristic title 'meester Gabriel' to refer to her teacher.

Note finally that this counterfactual analysis is quite restrictive. The narrator is not free to choose very vivid paraphrases, elaborate metaphors or similes (see 3.2) to characterize the perception but must stick with those words used in the the closest reflection-worlds, i.e. the most minimal verbal reflections.[7]

5 Mental stories

In Sections 2, 3 and 4, we've described characteristic linguistic features of narrated perception at the sentence level. But in the discussion of our main example narrative in Section 1, we also noted how narrated perception can be part of a larger stream of

7 A crucial difference with Stokke's modal analysis of character focus and protagonist projection is that our semantics of narrated perception (to be given in 7.4) involves marking the content as specifically perceptual. As noted in Section 1, not all examples of protagonist projection are perceptual in nature. As for the quotational part, the comparison with Stokke is less easy to make. Although his account with 'could' instead of 'would' seems less restrictive at first sight, his set of relevant worlds is further restricted to those worlds where the relevant character has the same beliefs and dispositions as in the fictional base world. A more detailed comparison would require the technical discussion of a number of philosophical issues beyond the scope of this paper.

consciousness representation, where what is perceived triggers memories, reflections, plans for the future etcetera (see also Banfield 1981: 74–75). This suggests that we should not just analyze the individual sentences of this stream in isolation. In this section we will illustrate the discourse-level semantic connections and dependencies within multi-sentence mental state descriptions and argue that narrated perception and free indirect discourse are best analyzed as constituents of a 'mental story', embedded inside the main story of plain narration. In Sections 6 and 7 we will then work out this idea in a general discourse semantics framework that combines SDRT (Asher & Lascarides 2003) with MSDRT (Kamp & Bende-Farkas 2006).

Consider again a fragment of our main example:

(30) Rosa looked at **her teacher**. **He** was writing. The fly was buzzing above **his** notebook. **Mr. Gabriel** waved it away, but the fly kept coming back. Apparently Hector wanted to know what **he** was writing.

Rosa's teacher, introduced in plain narration, is picked up by a pronoun in narrated perception. Subsequently this individual is reintroduced as 'Mr. Gabriel' in a narrated perception sentence, and then picked up by a pronoun in free indirect discourse. Modeling such ubiquitous anaphoric dependencies requires a dynamic framework where we keep track of discourse referents across sentences.

We should also pay attention to discourse coherence relations. As Bary & Maier (2014) argue for unembedded indirect discourse reports,[8] discourse coherence markers and inferences in multi-sentence report structures are problematic for static, sentence-level analyses that postulate silent intensional (or monstrous) report operators in the LFs of each individual sentence in a complex report. We can see the same thing here.

Take the discourse marker 'apparently' in (30). The perceived scene of Mr. Gabriel swatting at a fly that keeps coming back and the reflective thought of Hector wanting to know what the teacher is writing are causally connected in Rosa's mind. The marker 'apparently' marks this causal connection as an explanation, in Rosa's mental story: Hector's desire to know explains the fly's persistence (as Rosa is starting to realize the fly is actually a spying drone controlled by Hector). Crucially, the explanation relation is part of the mental story. It's not the narrator that comes to this realization marked by 'apparently', but Rosa.

To bring this point home, let's briefly consider what happens if we treat the narrated perception and free indirect discourse as separate sentence-level reports of different types, one with a perceptual operator and the other with a free indirect discourse operator. The problem with such an approach is that there would be no

8 See also more explicit arguments along these lines for free indirect discourse (Maier 2023) and dream reports (Maier 2024).

room for the explanation relation except at the level of plain narration. This leads to a reading where Rosa's seeing is, according to the narrator, explained by Rosa's thinking. This is quite different from the reading that we intend to capture in our dynamic mental story approach, namely: In Rosa's mental experience there's a perception of a fly that keeps coming back and a realization that this fly's behavior is due to Hector's desire to know what's being written. In other words, the narrator describes Rosa's complex mental experience as consisting of perceptions and thoughts, but including also the various anaphoric dependencies and coherence inferences between them.

To conclude, narrated perceptions often form part of a larger character's mental experience. Accounting for this feature is an important desideratum for any analysis of narrated perception. Together with the features that we have seen in the earlier sections, this provides the starting point for the analysis of mental storytelling (with both free indirect discourse and narrated perception) in the next sections.

6 A formal framework for modeling narrative discourse

6.1 Basic SDRT I: segmentation

On the SDRT model of discourse interpretation we start by segmenting the discourse we are to interpret, identifying the elementary discourse units, or EDUs, that express propositional content. Typically, these are the individual sentences, clauses, or nominalized verbs that plausibly introduce an eventuality into the domain of discourse.

Consider the following discourse:

(31) Sam opened the fridge. She looked for something to eat. She had been hungry all morning.

We segment as follows, identifying EDUs and labeling them with π_1 - π_3.

(32) π_1: Sam opened the fridge.
π_2: She looked for something to eat.
π_3: She had been hungry all morning.

Each individual segment in (32) indeed contributes one event or state discourse referent, and they also introduce (or anaphorically refer back to) individual discourse referents, as modeled by basic DRT. We'll use the notation in (33) to map a discourse segment to (a DRS representation of) its dynamic content.

(33) $\langle\!\langle \pi_1 \rangle\!\rangle = \begin{array}{|c|} \hline x_1 \quad y_1 \quad e_1 \\ \hline \text{sam}(x_1) \quad \text{fridge}(y_1) \\ e_1\text{:open}(x, y) \\ \hline \end{array}$

Some methodological notes are in order. First, since we are not concerned with compositionality, we will eliminate intermediate representations, take the model-theoretic dynamic semantics of DRT as a given, and content ourselves with specifying the interpretation of SDRT's discourse graphs as DRT boxes. In other words, strictly speaking (33) just maps a piece of an SDRT graph (an EDU label) to a DRT formula, which in turn has a semantic interpretation in terms of possible worlds, information states etc.. We defer the reader to DRT surveys like Geurts et al. (2016) for details of this last step.

Second, since we are also not concerned here with the syntax and compositional semantic interpretations of internal clause structure, presupposition, or anaphora, we greatly simplify our DRS representations, omitting the contributions of most verbal modifiers, including themes, subordinate clauses and sometimes even agents, from our boxes, leaving only the main eventuality and an abbreviated single predicate:

(34) $\langle\!\langle \pi_3 \rangle\!\rangle = \begin{array}{|c|} \hline e_3 \\ \hline e_3\text{:hungry} \\ \hline \end{array}$

6.2 Basic SDRT II: discourse graphs

When we read the text in (31) as a coherent discourse, we don't just compute the dynamic semantic contents of the individual discourse units, we also infer the existence of discourse relations between these units: Sam opens the fridge *and then* she looks for something to eat *because* she's hungry. SDRT introduces its so-called Glue Logic with defeasible axioms to model these inference processes (Asher & Lascarides 2003). The result is a discourse structure representation in the form of a graph, with discourse units as nodes, and the edges labeled with discourse relation labels. A common convention draws coordinating discourse relations (like NARRATION, moving the story forward) as horizontal arrows, and subordinating relations (like EXPLANATION, elaborating on a primary node) as vertical arrows.

(35)
$$\begin{array}{ccc} \pi_1 & \xrightarrow{\textsc{Narration}} & \pi_2 \\ & & \downarrow{\scriptstyle \textsc{Explanation}} \\ & & \pi_3 \end{array}$$

Following our methodological simplification we give the semantics of SDRT by showing how to map it to standard DRT. We do this by walking through the directed graph and interpreting each pair of discourse units connected by an arrow.

(36) $\langle\!\langle \alpha_i \rightarrow \alpha_j \rightarrow \ldots \rangle\!\rangle = \langle\!\langle \alpha_i \rightarrow \alpha_j \rangle\!\rangle \oplus \langle\!\langle \alpha_j \rightarrow \ldots \rangle\!\rangle$

The DRS representation of a pair of EDUs connected by a labeled arrow is determined by the semantic contribution of the discourse relation and the DRS representations of the EDUs:

(37) a. $\left\langle\!\!\left\langle \alpha \xrightarrow{\text{NARRATION}} \beta \right\rangle\!\!\right\rangle = \langle\!\langle \alpha \rangle\!\rangle \oplus \langle\!\langle \beta \rangle\!\rangle \oplus \boxed{e_\alpha \prec e_\beta)}$

b. $\left\langle\!\!\left\langle \begin{array}{c} \alpha \\ \big\downarrow \text{EXPLANATION} \\ \beta \end{array} \right\rangle\!\!\right\rangle = \langle\!\langle \alpha \rangle\!\rangle \oplus \langle\!\langle \beta \rangle\!\rangle \oplus \boxed{\text{cause}(e_\beta, e_\alpha)}$

When α and β are simple EDU labels,[9] their interpretations, $\langle\!\langle \alpha \rangle\!\rangle$ and $\langle\!\langle \beta \rangle\!\rangle$, are given by a compositional semantic construction algorithm mapping clauses to DRSs as illustrated in (33). The notation e_α in (37) is to pick out the main eventuality of the DRS $\langle\!\langle \alpha \rangle\!\rangle$. Finally, $e_\alpha \prec e_\beta$ means that the main eventuality contributed by α precedes that contributed by β.

Applied to the simple example in (31), we thus extract the following highly simplified, interpretable DRS from the SDRT graph in (35):

(38)
$$\begin{array}{|c|} \hline e_1 \;\; e_2 \;\; e_3 \\ \hline e_1\text{:open} \\ e_2\text{:look} \\ e_1 \prec e_2 \\ e_3\text{:hungry} \\ \text{cause}(e_3, e_2) \\ \hline \end{array}$$

9 The α and/or β in (37) may also be complex discourse units, drawn as unlabeled boxes in the graph. These boxes mainly serve as bracketing:

(i) $\left\langle\!\!\left\langle \boxed{\alpha_1 \rightarrow \alpha_2 \ldots} \right\rangle\!\!\right\rangle = \langle\!\langle \alpha_1 \rightarrow \alpha_2 \ldots \rangle\!\rangle$

A technical complication arises when a complex discourse unit consisting of multiple event-introducing EDUs is the argument of a discourse relation like EXPLANATION that is defined in terms of a main eventuality. We'll assume that the main eventuality can be created on the fly as the mereological sum of all top-level eventuality discourse referents.

This represents an information state with three eventualities: one of Sam opening the fridge, and one of looking for food, with the opening preceding the looking; and then one eventuality of being hungry, which causes the looking for food.

This concludes our introduction to SDRT. In the remainder of this section we add an account of reporting constructions in terms of the discourse relation of ATTRIBUTION, which will be extended in Section 7 to describe complex mental experiences.

6.3 Reporting, attribution, form, content, and characterization

We propose that the representations of subjective experiences and thoughts are uniformly represented under the non-veridical discourse relation of ATTRIBUTION (Hunter 2016, Maier 2023). In standard, overt indirect discourse speech and attitude reports this means we have a matrix clause segment introducing a speech or thought event, or some other mental state, and a complement clause segment representing the content of that event or state.

(39) a. discourse: Sam thinks she's the best

b. segmentation: π_1: Sam thinks $\quad$ π_2: (that) she's the best

c. unit representations: $\langle\!\langle \pi_1 \rangle\!\rangle = \begin{array}{|c|}\hline x \quad \mathrm{e}_1 \\ \hline \text{sam(x)} \\ e_1\text{:think(x)} \\ \hline \end{array} \qquad \langle\!\langle \pi_2 \rangle\!\rangle = \begin{array}{|c|}\hline \mathrm{e}_2 \\ \hline e_2\text{:best(x)} \\ \hline \end{array}$

d. discourse graph: $\begin{array}{l} \pi_1 \\ \big\downarrow \ \textsc{Attribution} \\ \pi_2 \end{array}$

Similarly, for direct quotation, the quotation itself is treated as a separate discourse unit. Following Maier (2023), quotation makes the form of the quotation relevant to the interpretation, which is captured by adding a phonological form as the first component of a form–content pair:

(40) a. Pip said "Sam is an idiot"

b. $\langle\!\langle \pi_2 \rangle\!\rangle = \left\langle \text{Sam is an idiot}, \begin{array}{|c|}\hline x \quad e_2 \\ \hline \text{sam}(x) \\ e_2\text{:idiot}(x) \\ \hline \end{array} \right\rangle$

The use of form–content pairs is important for our purposes since it is almost universally accepted in the growing linguistics literature on free indirect discourse that the interpretation of this construction is sensitive to the fine-grained linguistic form of the speech or thought representation, not just the semantic content as in regular indirect discourse (Schlenker 2004, Maier 2015b). We can represent free indirect discourse units as form–content pairs, where the form components have 'holes' for pronouns and tenses, represented with [unquotation brackets].

(41) a. π_4: Yes, today she was going to relax – finally.

b. $\langle\!\langle \pi_4 \rangle\!\rangle = \left\langle \text{Yes, today [she}_x\text{] [was}_t\text{] going to relax}, \begin{array}{|c|} \hline e_2 \\ \hline e_2\text{:relax}(x,t) \\ \hline \end{array} \right\rangle$

This simplified form–content pair captures the information that π_4 contributes its dynamic semantic content, viz. that there's an event of some previously established female x relaxing at time t, as well as parts of the linguistic form in which that content was originally tokened in thought: 'Yes, today [...] [...] going to relax', where the two holes stand for expressions that were originally used to (mentally) refer to the given female protagonist x and past time t, that is, probably, 'I' and 'am'.

The semantics of ATTRIBUTION must be sensitive to both form and content. For ease of exposition we introduce an auxiliary relation of characterization 'Char' (a two-place relation that gives rise to complex DRS conditions, relating a DRS to an event). The idea is that a DRS characterizes a thought iff that DRS accurately represents the semantic content of the agent's thought.

(42) $$\left\langle\!\!\left\langle \begin{array}{l} \alpha \\ \downarrow \textsc{Attribution} \\ \beta \end{array} \right\rangle\!\!\right\rangle = \langle\!\langle \alpha \rangle\!\rangle \oplus \boxed{\text{Char}(\langle\!\langle \beta \rangle\!\rangle, e_\alpha)}$$

Properly defining the model-theoretic semantics of Char-conditions in DRT, is not trivial, though the intuition behind it is relatively straightforward: a DRS K characterizes a contentful eventuality, say a thought, iff all worlds compatible with the thought are K-worlds. A more finegrained form–content pair $\langle \sigma, K \rangle$ characterizes a contentful eventuality, say a speech act, iff that same condition applies (all worlds compatible with the speech act are K-worlds) and in addition the surface form of that speech act is similar to σ in contextually relevant ways. Maier (2023) makes this a bit more precise, taking care of indexical shifts and such. He uses the functions *Content*, mapping a contentful event (like a thought or speech act) to a set of worlds (its content), and *Context*, mapping such an event to a Kaplanian context (centered on the world, time, and experiencer where the event occurs, Eckardt 2015):

(43) $[\![\text{Char}(\varphi, e)]\!]_w^{f,c}$ is defined iff $f(e)$ is a mental experience and φ is either a DRS K or a form–DRS pair $\langle \sigma, K \rangle$. If defined:

a. $[\![\text{Char}(K,e)]\!]_w^{f,c} = 1$ iff $Content(f(e)) = \lambda w.[\![K]\!]_w^{f,c}$
b. $[\![\text{Char}(\langle\sigma,K\rangle,e)]\!]_w^{f,c} = 1$ iff $Form(f(e)) = \sigma$ and $Content(f(e)) = \lambda w.[\![K]\!]_w^{f,Context(f(e))}$

Below, we extend this basic semantics of Char-conditions to cover the characterization of complex attitude representations with multiple form–content compartments.

7 Mental story attribution in SDRT

7.1 Complex mental experience

Let's take the following fragment from the story about Rosa:[10]

(44) Rosa looked at her teacher. He was writing. The fly was buzzing above his notebook. [. . .] Apparently Hector wanted to know what he was writing. Suddenly she knew for sure.

We segment as follows:

(45) π_1: Rosa looked at her teacher.
π_2: He was writing.
π_3: The fly was buzzing above his notebook.
π_4: Apparently, Hector wanted to know what he was writing.
π_5: Suddenly she knew for sure.

We read this as involving a mental story attribution, in which π_2 and π_3 describe Rosa's perception and π_4 describes a thought she has. To capture this reading, we need an ATTRIBUTION pointing to a complex discourse unit containing the mental story. The first complication is that there is no contentful mental state event described explicitly as taking place in the story world. Since ATTRIBUTION requires such a contentful eventuality, we will accommodate a segment introducing a mental experience, π'.[11] Presumably this mental experience is closely related to the event of looking at her teacher in π_1, i.e., she's looking at the teacher *and then* she has perceptual experiences and reflective thoughts. We connect π' to π_1 with NARRATION.

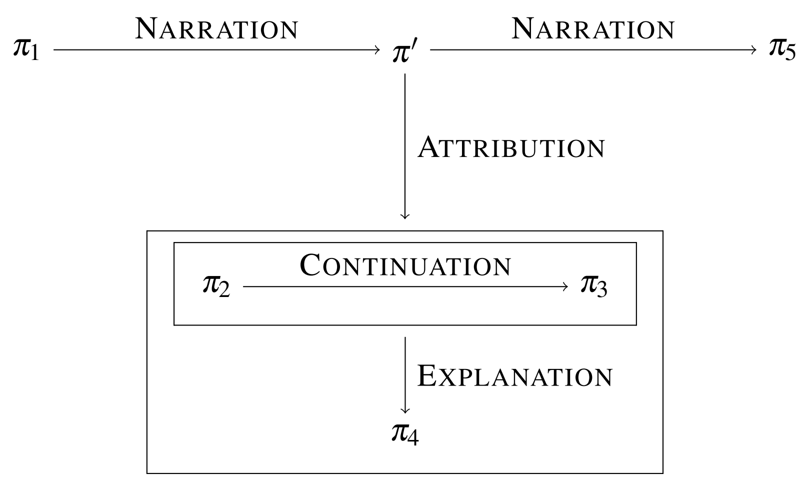
10 We're shortening the narrated perception by deleting one sentence from the original to keep the diagrams readable.

11 Maier (2023) makes a similar move for free indirect discourse, where a (often implicit) thought event is needed to capture the fact that free indirect discourse is a form of reporting. The current proposal subsumes that earlier account as a special case.

$$(46) \qquad \langle\!\langle \pi' \rangle\!\rangle = \begin{array}{|c|}\hline e' \\ \hline \text{mental.exp}(e') \\ \hline \end{array}$$

The following three segments flesh out what this mental episode e' is like. Through their form and content they tell the 'mental story' that 'characterizes' e'. In terms of discourse structure, they form a complex discourse unit connected via ATTRIBUTION to π'. Internally, π_2 and π_3 describe perceptions of the protagonist noting different aspects of the scene she's looking at, hence CONTINUATION. π_4 then describes an occurrent thought event by the protagonist, viz. a mental utterance of the form 'Apparently, Hector wants to know what he is writing'. Hector's desire to know what the teacher is writing is, according to Rosa, the explanation of the buzzing flies – as the flies are Hector's little spying drones. Hence π_4 is connected via EXPLANATION. Finally, π_5 brings us back to the narrator describing what's going on in the story world. That means π_5 must connect to π', again, via NARRATION.

(47)



The graph in (47) represents the discourse structure of the main storyline ($\pi_1 - \pi' - \pi_5$: she looks at her teacher, then has a mental experience, and then she knows for sure), and that of the mental story ($\pi_2 - \pi_3 - \pi_4$: teacher is writing and flies are buzzing, because Hector wants to know what the teacher is writing). The two storylines are linked by ATTRIBUTION.

7.2 Perceptions and thoughts as mental compartments

Inside the mental state box we have to capture the semantic contents associated with the discourse units that make up the mental story, making sure to capture the differences between free indirect discourse units and narrated perceptions. We'll take our cue from M(ental)S(tate)DRT, a DRT-based framework designed to represent the kinds of discourse dependencies observed in Section 5. MSDRT models complex mental states as consisting of various referentially dependent mental state compartments, i.e. anchors (mental files), beliefs, desires, imaginations, each represented as appropriately labeled DRS boxes (Kamp 1990, 2015, Maier 2015a).

A quick example to introduce the MSDRT framework. The complex mental state of someone who believes there's a monster under the bed and hopes it won't eat him looks something like this:

(48)
$$\boxed{\begin{array}{c} \left\langle BEL, \begin{array}{|c|}\hline \text{x} \\ \hline \text{monster(x)} \\ \hline \end{array} \right\rangle \\ \left\langle DES, \boxed{\neg\ \text{eat(x,i)}} \right\rangle \end{array}}$$

Such mental state representations can be model-theoretically interpreted (Kamp et al. 2003, Maier 2016) and integrated into a semantics of attitude report structures (Maier 2015a). We'll use compartmentalized mental state boxes like (48) as a way of making explicit the interpretation of ATTRIBUTION in complex mental story attributions.

In the cases we are currently interested in we must primarily separate (unreflective) perception from (reflective) occurrent thoughts, so we introduce the labels PERC(EPTION) and TH(OU)GHT. Moreover, as shown above, for free indirect discourse we represent not just the content of reported thoughts, but their linguistic form as well, with holes for (some) tenses and pronouns.[12]

(49) $\langle\!\langle \pi_4 \rangle\!\rangle =$
$$\left\langle THGHT, \begin{array}{l}\text{Apparently, Hector want[-ed]]} \\ \text{to know what he [was] writing}\end{array}, \begin{array}{|c|}\hline e_4 \quad x_4 \\ \hline \text{hector}(x_4) \\ e_4\text{: want.know}\ldots \\ \hline \end{array} \right\rangle$$

Unlike occurrent thoughts we can't assign actual perceptual experiences a language-like form. But, as observed in Section 4, word choice in represented perception may reflect the style, register, and word choices of the reported protagonist. Our descriptive generalization was that the words of the perception report must match with the words that the protagonist would have used if they had

12 Note that in the examples under discussion the third person pronouns all refer not to the attitude holder (Rosa), but to the teacher, who is a third person from the perspective of both Rosa and the narrator. Unquotation is therefore optional, or rather, it makes no discernable difference to the observable truth conditions in this context (Maier 2015b). We'll leave them as part of the quotation to mark that these 'he's do not pick out the experiencer of the mental state.

reflected on their perception. If we want to make this counterfactual quotation part of the truth conditions of perception reports we need the semantics (to be worked out in Section 7.4) to have access to the form of the report and hence we need to represent the PERC units as triples, just like with free indirect discourse. Note that tenses and pronouns in perception reports seem to behave exactly as in (free) indirect discourse (Brinton 1980, Abrusán 2021), i.e., they are generally unquoted.

(50) a. $\langle\!\langle \pi_2 \rangle\!\rangle = \left\langle PERC, \text{He [was] writing}, \begin{array}{|c|} \hline e_2 \\ \hline e_2\text{:write...} \\ \hline \end{array} \right\rangle$

b. $\langle\!\langle \pi_3 \rangle\!\rangle = \left\langle PERC, \text{ The fly [was] buzzing}, \begin{array}{|c|} \hline e_3 \\ \hline e_3\text{:buzz...} \\ \hline \end{array} \right\rangle$

7.3 From mental episode graph to MSDRS

We now have to extend our graph interpretation algorithm to deal with complex units that represent mental episodes, i.e., with EDU labels that get mapped to triples. If two connected units α and β inside a mental state box get mapped to triples our original definitions of coherence relations, no longer work as they (often) rely on DRS merge. We extend the definition of DRS merge $\oplus$ to triple with the same label Λ (i.e., either THGHT or PERC):

(51) $\langle \Lambda, \sigma_1, K_1 \rangle \oplus \langle \Lambda, \sigma_2, K_2 \rangle = \langle \Lambda, \sigma_1\sigma_2, K_1 \oplus K_2 \rangle$

But we've seen that perceptions may also be coherently linked to thoughts. In those cases we can't merge them under a single label, so we create a set in which they keep their own label:

(52) If $\Lambda_1 \neq \Lambda_2$, then: $\langle \Lambda_1, \sigma_1, K_1 \rangle \oplus \langle \Lambda_2, \sigma_2, K_2 \rangle = \begin{Bmatrix} \langle \Lambda_1, \sigma_1, K_1 \rangle \\ \langle \Lambda_2, \sigma_2, K_2 \rangle \end{Bmatrix}$

This effectively ensures that different types of mental states (perceptions and thoughts, as reported in narrated perception and free indirect discourse respectively) are represented as separate compartments. We'll use the familiar DRS merge symbol $\oplus$ to also denote set union:

(53) a. For two sets A and B, $A \oplus B = A \cup B$

b. For a set A and a DRS (or triple) K, $A \oplus K = A \cup \{K\}$.

As a result of generalizing the DRS merge operator, as it occurs in semantic definitions of discourse relations, whenever we have a mental state graph that contains EDUs that represent multiple distinct modes, i.e., thoughts and perceptions, we end up not with a DRS or triple, but with a set of labeled triples, which is essentially what an MSDRS is.

Applied to our example graph in (47) we thus derive the DRS representation of its truth conditions in (54), with the mental state graph under the ATTRIBUTION giving rise to an embedded MSDRS. Recall that the main eventuality of a complex unit, in this case that of the two perceptions, was defined as the sum of the top-level eventualities of the corresponding DRS, hence the causal relation introduced by interpreting the EXPLANATION relation relates the combination of teacher writing and fly buzzing to Hector's wanting to know. Note also that the causal relation is not itself labeled, so merging it with the perception and thought triples (as per the semantics of EXPLANATION) just means we add an unlabeled box to the set of mental state compartments. We'll interpret unmarked mental state compartments as representing a basic doxastic attitude.[13]

(54)
$$
\begin{array}{|c|}
\hline
e_1 \quad e' \quad e_5 \\
\hline
e_1\text{:look} \\
e'\text{:mental.exp} \\
e_1 \prec e' \\
\text{Char}\left(\left\{\begin{array}{l}
\left\langle PERC, \begin{array}{l}\text{He [was] writing} \\ \text{The fly [was] buzzing}\end{array}, \begin{array}{|c|}\hline e_2 \quad e_3 \\ \hline e_2\text{:write} \\ e_3\text{:buzz} \\ \text{continue}(e_2,e_3) \\ \hline\end{array}\right\rangle \\
\left\langle THGHT, \begin{array}{l}\text{Apparently he} \\ \text{want[ed] to} \\ \text{know} \ldots\end{array}, \begin{array}{|c|}\hline e_4 \\ \hline e_4\text{:want.know} \\ \hline\end{array}\right\rangle \\
\boxed{\text{cause}(e_4\ , e_2+e_3)}
\end{array}\right\}, e'\right) \\
\text{know.for.sure}(e_5) \quad e' \prec e_5 \\
\hline
\end{array}
$$

In words, there's an event of looking followed by a mental episode event e', which is characterized by the three conditions in the embedded box: (i) a perception: the teacher is writing and the fly is buzzing; (ii) a thought: apparently he wants to know what the teacher is writing; and (iii) a belief: those perceived events (of writing and buzzing) are the result of Hector's desire to know.

13 In Maier's (2015a) approach there's a fundamental belief-like attitude layer that all other compartments may (asymmetrically) depend on. In other MSDRT varieties we have internal anchors that ground the referents in other mental compartments and whose interpretation likewise depends on belief.

7.4 Interpreting complex attitudes in MSDRT

We're taking the semantics of standard DRT for granted, but the DRS that represents the truth conditions of our SDRS graph contains a few elements that go beyond standard DRT, viz. the Char(A,e) condition and its argument: A is a complex mental state representation, i.e., a set of labeled and unlabeled attitude compartments, and e is a mental episode, i.e., a complex event composed of perceptions, conscious thoughts, and beliefs in the mind of an agent at a time.

The general idea is that Char(A,e) is true iff (i) the form components of THGHT triples in A match the forms of thought-type sub-events of the entire mental episode e, (ii) the form components in PERC triples match the forms that perceptual sub-events in e would have if the agent would reflect on them, (iii) the DRS components in THGHT triples, PERC triples, and unlabeled doxastic compartments match the contents of relevant sub-eventualities of e. The definition of characterization in (55) below spells this out more precisely:

(55) $[\![\text{Char}(A,e)]\!]_w^{f,c} = 1$ iff there's a one-to-one mapping $m : A \to E$ s.t., for all $a \in A$, $m(a) \sqsubset f(e)$ and:
  a. if a is of the form $\langle THGHT, \sigma, K\rangle$:
    (i) $m(a)$ is a thought event and
    (ii) $Content(m(a)) = \lambda w.[\![K]\!]_w^{f,Context(m(a))}$ and
    (iii) $Form(m(a)) = \sigma$
  b. if a is of the form $\langle PERC, \sigma, K\rangle$:
    (i) $m(a)$ is a perception event and
    (ii) $Content(m(a)) = \lambda w.[\![K_1]\!]_w^{f,Context(m(a))}$, and
    (iii) if the agent of $m(a)$ were to reflect on $m(a)$, that reflective thought would have had form σ.[14]
  c. if a is just an unlabeled DRS:
    (i) $m(a)$ is a belief state and
    (ii) $Content(m(a)) = \lambda w.[\![a]\!]_w^{f,c}$

Note first that we will leave it open which semantics for counterfactuals would fit best ('if the agent were to reflect on their perception...').

Second, if we apply this definition of complex mental experience

14 If we assume a Lewis-style definition of a counterfactual operator $\Box\!\!\rightarrow$ (based on a relation of closeness between possible worlds, Lewis 1973), and $Reflect(x,e,e')$ means e' is an event of x (minimally, immediately) reflecting on experience e, and Ag maps an event to its agent or experiencer, we might, as a first approximation, attempt to precisify this counterfactual as:

(b-iii) $\forall e'.(Reflect(Ag(m(a)), m(a), e') \Box\!\!\rightarrow (Form(e') = \sigma)$

In words, those worlds where the immediate verbal act of reflecting on the experience has the stated form σ, are closer than those worlds where the agent's reflection has some other form.

characterization to work out the model-theoretic interpretation of concrete examples we'll run up against the main challenge faced by all accounts of complex attitudes, including MSDRT: on the one hand we need separate compartments to house separate attitudinal modes of a single agent's mental state, but on the other hand the various modes are referentially dependent on each other. One can believe there's a monster under the bed and hope it won't eat you, in which case the desire compartment is referentially dependent or 'parasitic' on the belief compartment, but not semantically compatible with it (as the experiencer, presumably, does not desire the existence of monsters under their bed).[15] Applied to the MSDRS in (54), the unspecified content compartment (contributed by the EXPLANATION discourse relation between the perceptions and the thought) is thus parasitic on the contents of both the thought and the perception. Since the underlying semantic issue of modeling parasitic attitudes is orthogonal to the matter at hand – narrated perception –, we defer to Maier (2016) for further discussion of parasitic attitudes and a general solution in MSDRT.

8 Conclusion

A good storyteller gives us a glimpse of the inner lives of their characters. In semantic studies of fiction it is commonly assumed that fictional narrators have access to their characters' innermost thoughts and share that knowledge with their readers through dedicated linguistic devices like free indirect discourse. Linguistically, free indirect discourse is a way of presenting a character's occurrent thoughts, blending direct quotation and indirect discourse. But not all of a person's mental experiences can be reduced to 'quotable' thoughts or inner speech. Perceptual experiences are a case in point. We have argued that there is a dedicated linguistic category of 'narrated perception', distinguishable from both plain narration and free indirect discourse. We have proposed a discourse semantics approach to narrative interpretation, where passages in narrated perception and free indirect discourse can join forces to present a fuller picture of the complex mental life of a fictional character.

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15 For discussions of similar issues and different solutions, see e.g. Ninan (2012), Maier (2015a), Blumberg (2018), Liefke (2023).

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